

Search sheets: how to run the activity



The pages after this one are 15 **search sheets**, built from *Green Eggs and Ham* by Dr Seuss. Print them one-sided and hand out **one sheet per person** — every sheet is different, and nothing needs cutting out.

If the room is a different size, match it to the sheets rather than the other way round. More people than sheets: print the set twice and hand out both copies. Every token pair is then held by two people, so every count doubles and the proportions — which is all the model is — do not move. It buys some insurance too, because one person away no longer takes a token pair out of the room with them. Fewer people than sheets: give the spares out rather than leaving them on the table, two or three to a person if it comes to that. A sheet nobody holds is a hole in the model.

Each sheet holds 66–67 **token pairs**. A **token** is one word or one punctuation mark; a token pair is a **next token** paired with the **previous token** that came immediately before it somewhere in the original text.

In **and ham**, the boxed token is the previous token and *ham* is the next token. Each token keeps the same colour wherever it appears, so scan by colour first and read second. With only 8 colours and far more tokens than that, a colour narrows the search rather than settling it — always check the token itself.

Say the colour when you call a token out — “who has *ham*? it’s a grey one” — and most of the page stops being worth reading. The swatches are chosen to answer to these words, so the words are the ones to use; the sheets themselves print no key, because a strip of swatches at the top of a page of swatches is one more thing to scan past.

black **grey** **red** **brown** **green** **blue** **purple** **magenta**

Running a round

Seed the board first. Ask someone to read out any token pair from their sheet — both tokens, boxed one first — and write it up. It matters that the opening comes off a sheet rather than out of your head: the room supplies the beginning as well as everything after it. Then, each round:

1. **Call out the last token** of the text so far.
2. Everyone scans their own sheet for token pairs whose previous token matches. **Hands up** — one hand each, and anyone with more than one match just goes with the first they spot.
3. **Pick a raised hand at random** and ask for that token pair’s next token.
4. A scribe writes the token on the board. That token becomes part of the context for the next round. Repeat.

Picking at random matters. If you take whoever shouts first you are sampling the fastest reader, not the text — and the whole point is that the room samples the way the model does.

What to watch for

The hands are the probability distribution. Every token pair was dealt to exactly one person, and the deal spreads each context across as many people as it will go, so a common continuation really does put more hands in the air than a rare one. Nobody has to explain weighted sampling; the room performs it. (Roughly: for a context more common than the class is big, somebody holds two matches but raises one hand, which flattens the busiest contexts.)

Nobody holds the model. No single sheet can continue the text on its own. The model only exists across the whole room — and if somebody is away, their share of it is missing, and some contexts will draw no hands at all.

Quiet rounds. One hand is not a stall — a word with a single continuation is the model working. Nor will you run dry: a word has nothing following it only if it appears nowhere but the very last position in the text, which is at most one word in the whole deal. So no hands at all almost always means an empty seat — two of them, if you printed the set twice — and that is worth saying out loud. Reseed from any token pair and carry on.

993 tokens, 56 unique · 1.8 bits/token · perplexity 3.5 · branching factor 3.45

Worked example

Start anywhere. Pick any token pair — say this one — and write its previous token *and* its next token on the board.

Round 1. Call out the last token on the board (*ham*). Another sheet has a match, so that person raises a hand and reads out ..

Round 2. Call out the last token on the board (.). Another sheet has a match, so that person raises a hand and reads out *Sam*.

Keep going for as long as you like. The text the room writes will sound like *Green Eggs and Ham* without ever repeating it — that is the model generating, not the corpus being read out.

the token pair that answers

and ham

ham .

! Sam

on the board

and ham

and ham .

and ham . Sam

and ham	and ham	do not	a mouse
would not	I will	train .	are so
would eat	!.	! in	! I
! Sam	them anywhere	a car	I am
here or	them in	them here	! I
could not	Sam I	not in	, could
not ,	not like	in the	them anywhere
let me	ham .	do not	dark ?
I will	like green	with a	a train
not like	I do	, Sam	am .
! I	eat them	, could	you may
? would	! I	you like	tree !
on a	! I	or there	you like
like them	box .	there ?	not ,
a train	will eat	them with	do not
goat .	the dark	in a	in a
green eggs	! I	I do	

in the	like them	, could	and I
them ,	, with	do not	them in
would not	a tree	would not	a train
do not	train !	I will	with a
you eat	there .	in the	and I
not like	tries them	not on	I will
you on	them with	eat them	? I
. try	green eggs	! If	a box
! a	. I	. not	he tries
eat green	here or	like green	eggs and
not ,	ham .	could not	house .
dark .	not like	not with	you do
I do	. and	. that	on a
or there	tree .	I will	Sam !
will not	. I	them here	am .
! I	with a	, will	do not
like them	in the	I do	

would not	train .	tree .	do not
green eggs	in a	them ,	do not
do not	a tree	I would	like them
. a	! so	or there	I do
, could	not eat	in a	eggs and
? I	would not	you !	I do
with a	them here	house .	could not
them ,	dark !	I will	like them
am Sam	Sam I	and I	them ,
. .	not like	not like	you do
. would	will not	, with	there .
with a	eat them	If you	and there
not ,	in a	. I	! would
. Sam	like them	on a	here and
! you	, Sam	mouse .	. I
not like	eat them	a house	you ,
a train	I do		

. and	tree .	train !	house ?
not eat	I would	would you	! Thank
, you	car !	. not	a house
am .	them anywhere	will eat	do not
, on	mouse .	. I	I would
. they	Sam I	them in	eat them
I do	them with	eggs and	green eggs
you eat	! I	not ,	eat them
not in	with a	a house	do not
you see	. say	say .	there .
a car	here or	like them	them in
in a	like them	with a	, Sam
would not	like them	Sam I	you ,
dark .	not like	Thank you	a box
not eat	in a	on a	I do
. you	I would	and ham	in a
? eat	could you		

them in	a box	not like	not like
not in	them ,	there .	I will
green eggs	I do	eggs and	a mouse
I would	not like	will eat	I would
in a	I do	. say	Sam !
anywhere .	Thank you	Sam ,	like them
here in	not like	in a	. I
with a	am !	would not	would you
like them	. would	them with	say !
do not	tree !	train !	eat them
a mouse	car !	house .	mouse .
. I	like green	, you	you may
. .	a boat	not in	with a
on a	! you	, Sam	do not
and ham	could not	. could	? not
! here	boat .	them !	eat them
you ,	dark ?		

in a	Sam I	a tree	I am
the dark	them anywhere	like them	car .
. I	will eat	train !	not like
boat .	you like	so like	. and
a car	like them	, could	Sam I
do not	not like	them anywhere	do not
eat them	not like	. I	am .
. I	tree !	you see	, in
eggs and	say .	. would	! not
house .	would not	not eat	them .
I would	may .	? not	. I
eat them	not ,	dark .	would you
in a	mouse ?	there .	them with
green eggs	a car	I do	with a
! a	not on	and ham	I will
on a	a tree	I would	like green
could you	anywhere .		

like them

like green

do not

I do

I will

! a

! I

. not

Sam I

do not

dark !

I do

boat ?

them !

not like

mouse .

a car

you ,

you may

in a

them here

in a

not like

not like

would you

could you

, could

green eggs

am !

eat them

. I

not in

car ?

ham !

house .

not like

eggs and

train ?

a fox

I am

and I

? I

anywhere .

them in

there .

. I

. I

I do

like green

may ,

. I

with a

so good

. .

the dark

a house

will not

say !

not in

them ,

eat them

would you

Sam I

I say

, on

a mouse

them with	? could	eggs and	not like
not like	anywhere .	! I	car .
not like	ham ?	not ,	would you
you ,	, in	in a	. I
mouse .	house .	say !	green eggs
will eat	. I	fox .	like them
I do	a box	them here	, with
I do	could not	a boat	with a
there .	a house	. I	and in
them .	am .	. I	may like
. I	you ,	I will	would you
! not	I like	like green	and in
. I	eat green	do not	them !
train !	so you	eat them	the rain
I will	. would	like green	I do
in the	do !	not ,	a train
not eat	Sam .		

I do	. not	in the	? I
, Sam	fox .	eat them	I am
there .	house .	would not	a box
not eat	you ,	with a	a boat
not like	am .	them in	. and
and you	eggs and	could not	rain ?
car !	do not	the rain	like them
not like	. you	I would	. not
not like	like that	them in	anywhere .
in a	Sam .	like green	would you
! try	will eat	eat them	try them
I am	not like	so good	I do
I am	may like	you ?	not in
. try	! I	mouse .	them ,
a goat	and ham	do not	them in
. I	, Sam	ham !	green eggs
a fox	. I		

. you	a tree	a fox	them .
eat them	a goat	the dark	will try
not in	them in	fox .	car .
here they	try them	eggs and	not in
, so	ham .	them with	you will
, in	I would	Sam I	. not
in a	not like	I do	in the
green eggs	! would	in a	eat them
not with	mouse .	like them	do not
. and	, I	? I	anywhere .
I do	. I	a fox	rain .
! Sam	with a	I am	be .
not ,	I am	would eat	you may
do not	. I	them here	not like
so I	box ?	would you	like them
and on	like them	I am	am Sam
and in	. I		

, in	, in	box .	not in
a house	I do	not in	am .
and ham	not like	! eat	with a
fox .	will eat	in a	try them
be !	eat them	am !	would you
rain .	anywhere !	them in	not ,
green eggs	and you	like them	I do
I am	. I	I do	ham .
or there	not like	them in	you let
them with	you ?	like them	? in
could you	. I	a goat	a train
do not	the dark	do not	I like
not like	. and	in a	here or
. not	eggs and	I will	. I
? I	a goat	in a	. and
would you	like them	, could	! Thank
Sam I	them anywhere		

will see	ham .	a fox	them in
be !	box .	. you	. and
or there	! you	could you	. I
not eat	! could	I am	fox .
not like	in a	them in	not eat
them !	try them	me be	. not
anywhere !	in the	like them	not in
. and	you let	I do	and ham
would not	eat them	a fox	? would
I will	, Sam	, Sam	. you
, would	am !	Sam I	do not
a house	the dark	and I	rain .
I do	eggs and	will not	do not
not ,	like them	a car	them ,
! and	with a	you ,	not with
I am	in a	here or	I do
see .	like them		

will not	not ,	do not	in a
in the	I will	I will	the dark
them with	a house	them and	see .
eat them	. not	ham ?	! let
not like	me be	. I	not ,
they are	a train	! that	a fox
goat !	, would	be ,	that Sam
let me	like them	! and	and ham
you ,	with a	Sam I	would not
them .	in a	I would	here or
am .	like them	. I	or there
. so	do so	will not	them with
I would	. I	not ,	could not
? I	like them	not ,	a mouse
do not	you ,	I am	. not
fox .	, could	and ham	, could
I am	box .		

not eat	them with	or there	, Sam
could you	a tree	I would	not like
box .	you like	, would	not like
goat ?	! I	I do	here or
ham .	. I	I am	? here
like them	the rain	like them	like them
not with	. I	fox ?	! would
. would	. and	a mouse	they are
that Sam	let me	you let	them ,
in a	see .	me be	I will
. say	Sam I	and ham	not eat
them anywhere	them ,	a mouse	I do
will not	. not	and I	do not
would not	a box	will not	do not
do not	in a	in the	good ,
I do	! I	eat them	with a
am .	, Sam		

let me	not in	them here	I am
a box	I would	. and	will not
and he	I do	in a	you say
you will	like them	would not	. I
goat .	I could	I do	do not
will see	! I	them anywhere	a train
not in	with a	, Sam	in the
, could	good ,	them here	are .
not with	. not	do not	! not
see !	I do	would you	box .
am .	. .	and I	a car
here or	do not	? I	ham !
like them	not eat	. not	eat them
! not	them here	, I	or there
a mouse	Sam I	that Sam	not in
the rain	in a	like them	could not
. I	me be		